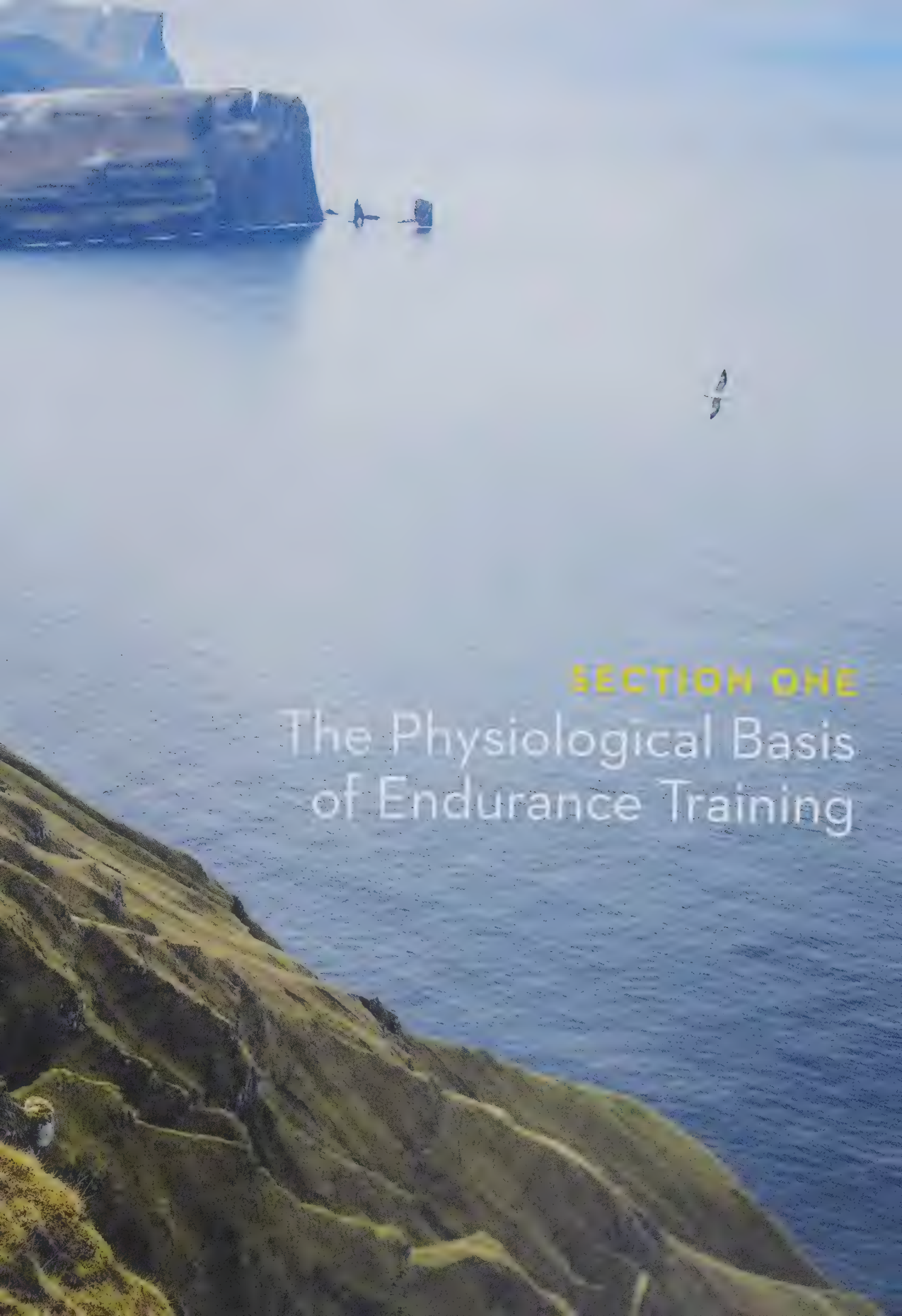




SECTION ONE

The Physiological Basis of Endurance Training



CHAPTER TWO

The Physiology of Endurance

“All models are wrong. Some are useful.”

– George Box, mathematician

Please don't rush past these important pages in your hurry to see what workouts you need to do to beat Kilian in his next race. You won't find that info. As we say numerous times throughout this book: There is no single recipe for maximizing fitness. Only you can optimize the ideal plan for yourself. We have included these early chapters exactly because of this individualistic nature of training. We *know* that you need to have this information to make good decisions about your training plans.

In this chapter we will take a look at the physiology underlying the production of the energy that enables endurance exercise. It will help to have a basic understanding of how this physiology works so that you have an intellectual framework for making decisions when it comes to designing and implementing your own or others' training. Blindly following any training scheme without understanding the underlying principles will handicap, if not undermine, your performance. What follows is a look at the physiology of endurance.

The Evolution of Endurance

For thousands of years, several coevolving protohuman species existed alongside of and competed with *Homo sapiens* as hunter-gatherers. The fact that we won

Previous page: Luke Nelson runs along the northern reaches of Eysturoy Island, Faroe Islands, Denmark. Photo: Kelvin Trautman

Opposite: Kilian Jornet running in the mountains above Åndalsnes, Romsdal, Norway, two weeks before heading to Cho Oyu and Everest in March 2017. Photo: Jordi Saragossa

the evolutionary contest is partly due to our species' endurance capacity. Scarcity was the normal state of affairs and competing for and obtaining calories required a lot of time and a lot of calories, so protoman's metabolism evolved to give those with the most economical use of those precious calories the best chance to pass their genes on to the next generation.

Most days involved many hours of low-intensity searching for food with occasional short bursts of high-intensity effort sprinkled in during the heat of the hunt or when the need arose to overpower a targeted prey or out-compete another predator or scavenger. The components of the early hominid diet were high in animal fats and protein, with a smattering of complex carbohydrates from plant sources. Early man's physiology developed a caloric fuel tank in the form of fat stores (both intramuscular and subcutaneous) for the excess calories consumed during times of plenty. Fat could sustain our ancestors on those long days of hunting and gathering as well as give a greater chance for survival in times of food shortages. We evolved to store fat in vast quantities to last many hungry days before incurring permanent damage to our bodies. The well-adapted hunter-gatherer could go longer between meals and operate at higher intensities on fats, thus conserving the scarce and precious stores of glycogen that came from carbohydrates. Another interesting aspect of this topic is our ability to recover muscle glycogen stores quickly during periods of rest (e.g., hours instead of days). This is not the case for most other animal species; thus, we could run our prey into the ground since we needed only a short period of time to recover.

Fat is a complex molecule with many chemical bonds, each bond containing usable energy. Even a lean, well-trained endurance athlete can carry up to 100,000 calories stored as readily accessible fat. Carbohydrates, being simpler molecules with less energetic bonds, yield about half the calories per gram than fat does. We have a much smaller carbohydrate-storage capacity of no more than 2,000 calories, even in a well-trained athlete. Our body's strategy during times of excess is to store any calories we consume, from whatever source, as fat. On a purely physiological level, this explains the modern-day obesity epidemic; many humans consume an abundance of calorie-dense foods and are rarely exposed to food shortages.

According to one popular theory of evolutionary biology, early hominids exploited their endurance and lack of hair (which allowed them to avoid the overheating fate of their prey) to run their next meal to the point of exhaustion. This helped them rise to the top of the food chain despite their relative physical weakness. It was by virtue of their endurance that our predecessors could get close enough to kill an otherwise much more powerful animal once the prey became too tired to defend itself. The theory goes on to say that the high-protein diet resulting from successful hunts allowed increased brain capacity and complexity and consequently, a cognitive revolution, which in turn led to all the subsequent cultural advances and evolutionary traits we have inherited. It follows then that

you can read the words on this page precisely because our species survived by utilizing its inherent ability to sustain a long duration of low- to moderate-intensity work. We are the product of an evolutionary heritage that has predisposed us to endurance.

(Liedenberg, 2008; Billat, et al., 2003).

Endurance and Fatigue

Endurance training is aimed at improving our ability to run, climb, or ski for an extended period of time. Endurance is ultimately limited by our bodies' predictable fatigue response to these activities. It is fatigue that limits endurance. Because of this, a brief discussion of fatigue is in order.

Within athletics, endurance is the maximum sustainable work rate (e.g., speed or power) an athlete can maintain for the duration of an event before fatigue dictates a reduction of the work rate. Fatigue in our sports is manifested as reduced stride length and slower stride rate. Several interconnected physiological systems work together to dictate endurance performance in events of differing durations and intensities. For example, a vertical-kilometer (VK) race is run at a very different intensity than a 50K trail run. However, both events will still test whatever the specific endurance/fatigue limit of the runner allows. The type of endurance needed and the type of fatigue experienced are relative to the specific event.

We don't need an exercise scientist to tell us that fatigue causes us to slow-down. Proper training makes you more fatigue resistant and thus forestalls the slowdown. We are very complex organisms and multiple adaptations to several of our bodies' systems are required to improve our resistance to fatigue. Simplifying: that dreaded reduction in speed that we all loathe is caused principally by 1) your body's inability to meet the energy demands of the exercise. This is caused by 2) a depletion or an accumulation of certain metabolites, or 3) a reduced motor nerve signal.

We can group these several physiological systems, roughly, as you see below. What we are doing is creating a model, an artificial delineation and segregation of these systems that are, in reality, intimately interconnected and interdependent. This simplified model format is commonly used in science to allow complex systems and ideas to be broken into their constituents, which may then be better understood. The art of coaching comes, in part, from an understanding of the interconnectedness and interdependence of these systems.

The Brain and Central Nervous System

The brain is the master control. It exists in a complex web of feedback and feed-forward messages. The Central Governor Theory, developed by Tim Noakes, MD, PhD, postulates that the brain is the ultimate arbiter of fatigue and regulator of

endurance. Despite not being able to physically locate the “governor” inside the brain, this model currently does the best job of explaining all of the limitations of endurance performance. In brief it states that, no matter the source or type of fatigue, the response of the brain is always the same: a reduction of muscle activation (reduced power output), causing you to slow down. The theory goes on to say (and some evidence suggests) that training results in raising of the governor’s limit so more work can be done before the state of alarm is reached and the governor kicks in.

The Oxygen-Delivery System

The heart, lungs, and blood vessels make up the oxygen-delivery system responsible for supplying oxygen (O_2) to all the cells of the body, including our primary focus: the working skeletal muscles.

Lungs. While we may feel as if we are “out of breath” during strenuous exercise, healthy lungs are actually overbuilt for our needs and have more than adequate gas-exchange capabilities. Human lungs have the surface area of one side of a tennis court.

Heart. From an O_2 -delivery standpoint, many scientific investigations have shown that the heart’s pumping capacity is the biggest limiter to O_2 delivery in healthy people. However, while the lungs do not adapt after puberty, the heart *can be trained to increase its output for each beat*, which means an increased output of O_2 and improved O_2 delivery to the muscles. This adaptability is limited by your genetics and history of endurance training. The heart reaches full maturation during adolescence, adapting fastest and to the greatest extent during this period. Young, as well as older, untrained individuals can see rapid and significant changes in the heart’s stroke volume because the heart muscle is very trainable; but, only up to a genetically predetermined limit. This stroke volume limitation ultimately sets the absolute upper limit for O_2 delivery and hence becomes a limit to aerobic power output. Athletically mature individuals (those with long endurance training history) will see small, if any, changes in stroke volume due to training.

Oxygen-Utilization System (Aerobic Metabolic System)

The rate at which your muscles can utilize the O_2 that is delivered to them is another primary determinant of endurance performance. This capacity for O_2 utilization is also the most highly trainable system in the endurance arsenal. It is dependent on the aerobic qualities of the muscles, which can be improved for many years, well into athletic maturity. We’ll dig pretty deep into this vital component later in this chapter, so for now, remember that this is the main focus

of most endurance training and, to a very large extent, endurance is the consequence of upgrading the aerobic metabolic system with the application of the correct training stimuli.

Muscular System

This is where we can no longer ignore the interplay of the above systems. The brain, through the motor nervous system, controls which muscle fibers are recruited and the duration of their activation. As mentioned above, the particular metabolic qualities that those muscles are adapted for (or not adapted for) play a commanding role in endurance. Different training practices influence which muscle fibers are recruited to do the work and the training stimulus those fibers receive. Long-duration training improves the aerobic qualities primarily of the slow-twitch fibers (ST). Shorter, high-intensity training improves the anaerobic (or glycolytic) qualities primarily of the fast-twitch fibers (FT).

Fatigue resistance is the desired outcome of endurance training. Effective endurance training works by causing fatigue to the systems described above. This in turn induces adaptations to occur with these systems. Because these systems are interconnected, monodirectional training (e.g., training primarily one system) will, after some time, inevitably yield diminishing returns. An athlete experiences this as a plateau, and in the worst cases, a degradation, in performance.

The Factors That Determine Endurance Performance

Many factors come together to determine how you will perform in any endurance situation.

Conventional wisdom and research give credence to the notion that endurance performance is supported on a tripod of three main attributes:

- 1) VO_2 Max: VO_2 max is defined as the maximum volume of oxygen you are capable of taking up and utilizing during intense exercise. This quantity is measured in milliliters of oxygen used in a minute divided by the body weight (in kilograms). This gives an aerobic power-to-weight ratio and is written as ml/kg/min. This quality is one that responds best to training in the young or the undertrained and can be difficult to change in the well trained. See the “Misunderstanding VO_2 Max” sidebar on pages 56 and 57.

VO_2 max is the product of the heart’s output and oxygen utilization of the skeletal muscles. Increasing the working muscle mass results in more oxygen use. Sports like cross-country skiing and skimo, which use all four limbs for propulsion, place a big demand for oxygen delivery from the heart. This results in skiers often having very high VO_2 max measurements. Untrained men and women



ATHLETE STORIES

Making the Decision

Janelle Smiley

My husband, Mark, and I were walking along a valley road in the Alps, twenty-one days into what would ultimately be a thirty-six-day ski traverse of the range. Of the three women who'd started out as part of our international team, I was the last remaining.

I looked over at Mark. "Do you realize that if I finish this, I'll be the first woman to cross the Alps on skis?"

He lit up with excitement. "Janelle, we are doing this!" he exclaimed. Right then, I made the decision that quitting was not an option. I was going to keep touring unless something maimed me. Before that moment, I was just trying to survive from one day to the next. Now, for the first time all trip, reaching the goal felt real.

Earlier that winter, Mark and I had been invited to join a group of seven athletes—a mix of alpinists and skimo racers—to ski from Vienna, Austria, to Nice, France. It seemed like a dream come true: to be on my skis for up to forty days crossing an entire range.

Ski touring is by far my favorite thing to do, and this big objective would call on both my racing and climbing backgrounds. It was a challenge beyond my ability and beyond what I thought was possible. In my heart I wanted it, but I wasn't convinced I could do it. I'd never gone more than seven days without a rest day, and now I'd hammer for five to six weeks without rest. Could my body handle the workload?

We met our five teammates for the first time during a one-day orientation in January. There were three men—an Austrian, a German, and a Swiss, who were skimo racers—and two women, a Spanish runner and an Italian alpinist. Alpinists and racers don't

necessarily look at the mountains with the same mindset; where racers are all about speed, alpinists tend to be about teamwork and managing terrain with an eye for safety. Mark's more of an alpinist, and I straddle the two disciplines.

The team dynamics were tense from the get-go, when the skimo boys—all German speakers—took off from the start line at race pace. As a group, we'd agreed to break the route planning into four sections and then share those plans in advance, which they neglected to do. Two of the skimo boys had planned the first section of the route with race pace in mind. So when they headed out at top speed, speaking a language I didn't understand, I was flying blind. I couldn't take part in any critical decision-making, and I didn't know where I was going except for a line I'd received the night before on my GPS.

The first week was chaotic—a series of ten- to fifteen-hour days spent chasing the racers in white-out conditions and high winds. If you had to pee or change a layer, they were gone; if you were to fall into a crevasse or break a ski, they'd already be halfway down the mountain. I couldn't do much more than get through each day.

I felt out of control, but I didn't want to quit. I kept going out of curiosity; I wondered when I would break, either mentally or physically. To help manage the mental component, I shifted my mindset to gratitude. I had to find something to be grateful for, otherwise my mind would take me down a destructive hole and I didn't have the mental capacity to go there with so many external stressors.

Janelle Smiley, somewhere in France ... or maybe Italy ... on day twenty-nine of Der Lange Weg (The Long Trail), an iconic ski tour that's reputed to be the world's longest, traversing the Alps from Vienna, Austria, to the beach in Nice, France. The team Janelle was part of succeeded in breaking the forty-day record—set in 1971—by more than three days. Photo: Mark Smiley



Philipp Reiter crossing underneath the north face of the Matterhorn on the biggest day (74 kilometers and 4,500 meters of climbing) of Der Lange Weg 2018 during a significant storm. Photo: Mark Smiley

The group broke up twice that week over differences in attitude and approach, especially toward risk tolerance, but each time we were able to stitch ourselves back together in the interest of pushing on.

Our numbers did eventually dwindle: First the Spanish runner dropped out because she was uncomfortable with the style of travel and its associated dangers. Later the Italian alpinist withdrew due to an overuse injury.

My mental fatigue throughout those first seven days was overwhelming. I'd take my boots off at the end of each day and wonder how I was going to get up the next day and put them on again. Then I'd get up the next morning, decide to put my boots on, and start moving. After a few steps, my body knew what to do and my mind followed along. By the eighth day my body had grown used to the routine of the workload.

We covered so much ground—up mountains, down mountains, through valleys that sometimes expanded, sometimes narrowed. Clouds swirled and danced around the peaks. As the days inched by, I felt my mind quieting. On grades that were perfect for a gentle downhill glide, slow and controlled as if I were on an escalator, I'd watch the mountains go by and do nothing but enjoy their beauty. I fell into trances looking at the snow passing underneath my skis.

I decided to surrender myself to the circumstances—to these strangers in our group, to the mountains, to the variable conditions and harsh weather. I decided to *like* things that were otherwise uncomfortable or

frustrating. Instead of turning my back to the wind, I faced into it, making me feel alive. I developed a giddy appreciation for the warmth of my protective layers while in a whiteout. I chose to be thankful for an abrasive teammate; if it hadn't been for his constant pushing, I might never have glimpsed what's possible inside this body and this mind. I continually made the decision to find the good in each situation. I had to—it's what carried me through.

On day twenty-one, when I made the decision to complete the traverse, my curiosity to see if I could do it turned into belief that I would. Fifteen days later, our now-five-person team walked into Nice, skis strapped to our packs. We'd averaged about 35 miles and 8,500 vertical feet each day.

This traverse of the Alps magnified my thinking about how our decisions dictate our outcomes. I still can't believe we pulled it off, but I'd decided I was going to finish, and with that decision came success. ■

Janelle Smiley grew up in Aspen, Colorado, and lives in Jackson Hole, Wyoming. She is a six-time ski mountaineering champion; she has completed forty-eight of the fifty classic climbs of North America; and in 2018 she became the first woman to cross the Alps on skis. She currently works as a mindset performance coach, helping individuals bring body, spirit, mind, and heart into alignment toward performance goals. Learn more at janellesmiley.com.

typically show values in the range of 35–40 ml/kg/min and 25–30 ml/kg/min, respectively. The highest VO_2 max values for males are in the range of 90–95 ml/kg/min. Females are generally lower due to a lower percentage of lean muscle mass to whole body weight and top out at about 80. To put our aerobic power into context: Thoroughbred race horses usually have about 180 ml/kg/min. Top-flight racing dogs will exceed 250 ml/kg/min. Hummingbirds come in at over 600 ml/kg/min and the winner is the honey bee with the equivalent of over 6,000 ml/kg/min. Recall our early reference to our relative weakness in the animal kingdom.

2) Movement Economy: Just like a car's fuel economy, this is the measure of how much energy it costs you to move your body a certain distance. And, like your car, it is specifically related to the energy used to move you at a certain speed (race pace). Intuitively you can see that if athlete A uses 20 percent more energy than athlete B to cover one kilometer in four minutes, then A will need more endurance to sustain that energy output or will tire more quickly. Movement economy combines two components: the mechanical (technique) and the metabolic. Good technique implies an efficient use of the energy needed to move at a given speed. Less energy is wasted compared to inefficient movements. Watch the graceful, floating running style of a sub-2:10 marathoner compared to the plodding of the four-hour competitor. This is technical economy. Metabolic economy, the second component, while invisible to our eyes, is equally as important. Metabolic economy is dictated by the metabolic pathway our muscles use to produce the energy required to sustain the exercise intensity. As we dive into the metabolic systems in the next few pages, you will see that not all metabolism has the same energetic costs or creates the same stress on the system. Keep in mind that both the technical and metabolic components of economy are highly trainable.

3) Lactate Threshold: Also known by the confusing array of names listed below* (thank you, exercise science!). We will discuss Lactate Threshold (LT) in much more detail soon, but for now understand that this is the maximum work rate you can sustain for a long duration. If that sounds like endurance to you, then you're correct. The pace at LT is the best predictor of endurance performance. The rate at which your metabolism can sustainably produce energy is the single largest determinant of your endurance. Luckily, this metabolic effect is perhaps the most adaptable to training of the endurance tripod. Because of this, we're going to roll up our sleeves and dive right in.

**They are, in no particular order, Anaerobic Threshold, Functional Threshold Power or Pace, Critical Pace, Critical Speed, Critical Power, and Second Ventilatory Threshold. While the meaning and even existence of these terms engender serious scientific dispute, we lump them here as having the same basic meaning to the coach and athlete. They represent the maximum pace sustainable for an extended (30–60 minutes') duration.*

Endurance Seen Through the Metabolic Lens

We're going to deal first with the last of the above-mentioned three legs of the performance tripod. We will return to the others shortly.

In a healthy and well-trained individual engaging in events lasting less than two hours, endurance is limited primarily by the rate the muscle cells can sustainably produce the energy required for their contractions. For events lasting longer than two hours, fuel stores become the major limiter (we'll cover that soon). Training for endurance then must involve optimally organizing and executing a long-term series of workouts to increase the muscles' sustainable metabolic output. Keep this basic idea in mind as we move through the book and it will help frame why smart training needs to be targeted at enhancing specific metabolic processes rather than the popular, but misguided, method of going as hard as you can, for as long as you can, as often as you can.

ATP's Role in Endurance

What enables endurance? More specifically: What powers the muscular work that propels you up the mountain? The answer can be stated in three letters: ATP.

Adenosine Triphosphate is a short-term energy storage molecule used by every cell in your body. If any one molecule can be singled out for its life-giving force, ATP is it. The energy released when an ATP molecule's chemical bonds are broken is what powers all the functions of cellular life.

Most importantly for athletes, ATP's energy is used to power muscular contractions. The more work you do, the more ATP you use. The ATP supply, specifically its rate of production, thereby becomes the limiter of the speed you can run for durations exceeding a few seconds. Endurance athletes need to not only improve the rate of metabolic turnover of ATP but, at the same time, they need to ensure that these higher rates can be sustained for as long as needed for their specific event.

Metabolism: The Energy Transfer from Food to Physical Work

The energy in the food you eat is not directly available to the body. The digestive process first breaks down the food into its macro nutrients: fat, carbohydrate, and protein. These fats and carbohydrates are further converted into various fatty acids and sugars that can then be used to produce ATP.

Metabolism is the term for the biochemical process of breaking down the molecular bonds in the fats and sugars for the energy needed to assemble the molecule ATP. Think of metabolism as a miniature ATP recycling plant in the





muscle cells. After ATP is broken apart to extract the energy needed for muscle contraction, then the metabolic recycling plant using the energy from food reassembles the ATP molecules so the whole process can keep going.

It's a good thing we recycle ATP because it is a heavy molecule. The amount of ATP required to sustain a resting metabolic rate for one day would amount to about 60–70 kilograms. Basically an extra “you” if we had to store and carry all this ATP.

THE TWO METABOLIC PATHWAYS TO ATP PRODUCTION

We rely on two unique metabolic pathways to create ATP. Each process involves multistep chemical reactions. Each pathway yields a different quantity of ATP per unit of the fuel (carbohydrates and fat) consumed. Each pathway can change its rate to meet energy demands up to its maximum limit or capacity. That capacity is what we aim to increase with training.

Anaerobic Glycolysis Metabolic Pathway

This is the method by which carbohydrates fuel you. All the various sugars contained in carbohydrates are converted to glucose by the liver and either used directly or stored in the muscles or liver as glycogen for later use. Anaerobic glycolysis is a ten-step process of breaking apart the glycogen molecule. This process does not require oxygen (hence the modifier Anaerobic). The end products of anaerobic glycolysis are a molecule called pyruvate along with two ATP molecules. This ATP can be directly put to use for energy. But it is the fate of that pyruvate that is central to the discussion of endurance, so we'll come back to it after a look at the other metabolic pathway we use to make ATP. Glycolysis can proceed at a much faster rate than the aerobic process (following) so is used when ATP demand is high and exceeds what can be supplied with aerobic metabolism such as during high-intensity exercise like running very fast.

Aerobic Metabolism

This complex, multistep chemical process occurs inside the muscle cell's mitochondria (see sidebar on page 37) and requires oxygen in order to function. It can use those pyruvate molecules produced by the anaerobic metabolic path mentioned above as well as fat to produce additional ATP. Aerobic metabolism produces roughly seventeen times the amount of ATP that the anaerobic path makes but is a more complex and hence slower process. The relative amounts of fat and sugar that contribute to its prodigious ATP output will become central to our understanding of how training methods affect endurance*. For many years it was thought that, due to its slow rate of ATP synthesis, the aerobic metabolism was incapable of meeting

the demands of high-energy outputs. Recent studies on elite endurance athletes have shown that the aerobic contribution can be significant right up to the highest intensities in well-trained and fat-adapted athletes. More on this later.

**Aerobic metabolism can utilize protein to produce energy. But protein contributes only a tiny fraction of the energy used for exercise.*

Mitochondria

Mitochondria are tiny organelles in every cell of your body (besides blood cells) and are considered the powerhouse of the cell. These tiny factories use oxygen in the air you breathe to produce ATP, the essential fuel of life in all the cells of your body. Slow-twitch muscle fibers have a high concentration of mitochondria, predisposing

them to aerobic metabolism. This makes these muscles more fatigue resistant than their fast-twitch neighbors. Fast-twitch muscles are less aerobically endowed and have to rely more upon anaerobic metabolism to produce their power.

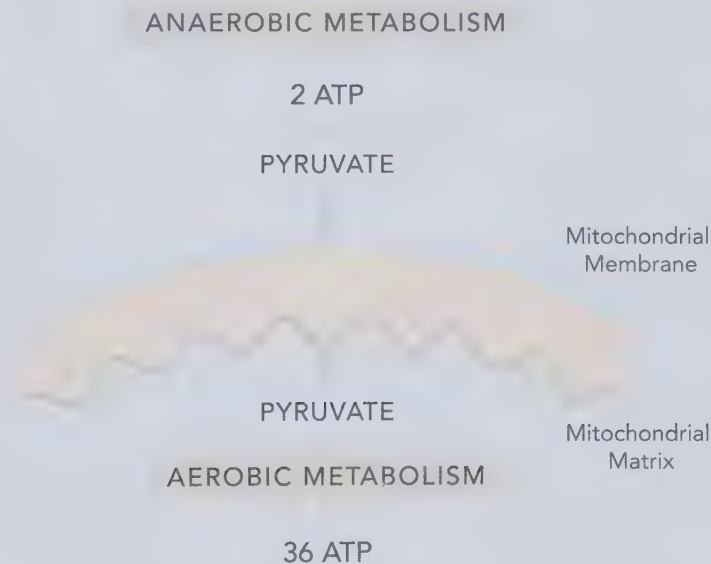


Fig. 2.1 Mitochondria and the Fate of Pyruvate

Pyruvate, the end product of anaerobic metabolism (glycolysis), can either linger in the cytosol of the muscle cell—a metabolic dead end—or it can be transported into the mitochondria where it undergoes aerobic metabolism, which produces a great deal more ATP (energy for muscle work). Mitochondria are the powerhouses of life in all our cells.

The Role of Metabolism in Endurance

The connection between muscle contractions (mechanical work), ATP (chemical energy), and metabolism can now be made more clearly:

- ATP fuels muscle work and is consumed in the process.
- Metabolism is the process whereby ATP is resynthesized.
- The faster ATP is recycled, the more work you can do per second.
- Work per second is defined as power.
- More ATP = more muscle power.
- More muscle power = more speed (all things being equal).

At its most elemental, then: Endurance is the ability to sustain rapid ATP production, predominately through aerobic means, for long durations. When an athlete improves endurance, he or she is able to maintain higher speeds for a longer time. This is how we described the quality “endurance” earlier. It should make more sense to you now.

Because endurance training should enhance the athlete’s ability to produce greater speed for longer periods, it follows that endurance training improves the athlete’s ability to sustain a high rate of ATP turnover. With this

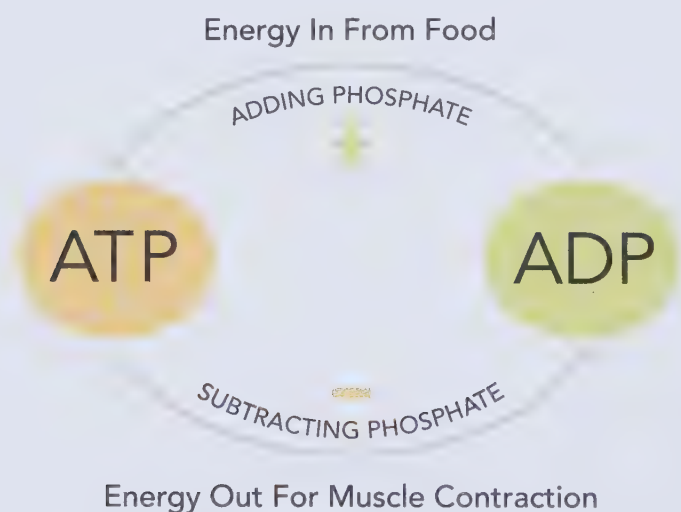


Fig. 2.2 The ATP Recycling Plant

Metabolism functions as a recycling plant. The energy contained in the food you eat is used to recombine the severed phosphate molecule with ADP (adenosine diphosphate) making ATP again. The cleaving of this third phosphate from ATP provides the energy of life—and the energy to run or ski. This continual recycling of ATP, to ADP, and back to ATP is called metabolism. If we couldn’t recycle ATP, we’d need massive ATP stores (tens of kilograms) just to get through a normal day let alone one filled with heavy exercise.

ALL Endurance Training Enhances Metabolic Turnover



Fig. 2.3 More ATP Equals More Power

The role of all endurance training is to increase the output of the ATP recycling process. The faster your metabolism can recycle ATP, the faster you can run.

understanding of the relationship between metabolism and endurance we can focus our attention more closely on how we can manipulate our metabolism's role in endurance through training.

Now you understand why we say: *Endurance is a metabolic quality.*

Lactate Dynamics, the Key to Endurance

You are much closer to understanding the basis of how you will make coaching and training decisions in the future. You understand why our metabolism determines our endurance. The two metabolic pathways differ in distinct ways. The next several pages will explain these distinctions. This will help you understand how and why you must apply the correct training stimulus to get the desired training effect.

Metabolic Mix

Whether sitting still or running a race, both the anaerobic and aerobic metabolic engines are contributing to your body's energy needs. The relative contribution from each is a result of several factors:

- **Intensity** – As intensity (speed) increases, so does the demand for ATP. At a certain speed, the aerobic system alone will no longer be able to meet the ATP demands and ATP from glycolysis will begin to dominate overall energy output.
- **Genetics** – Those naturally endowed with more slow twitch muscle fibers will be able to produce more ATP aerobically as speed increases and delay the dominance of glycolysis.
- **Training history** – Those with a history of high-volume, low- to moderate-intensity training will have slow twitch muscles that are aerobically well

adapted and be able to sustain higher speeds aerobically. Likewise, those with a history of higher intensity will have predisposed their metabolism toward the anaerobic/carbohydrates use.

Lactic Acid: A Metabolic Fork in the Road

Let's revisit the pyruvate molecule that is the end result of anaerobic glycolysis. Pyruvate can take one of two roads: It can enter the muscle cell's mitochondria and undergo additional aerobic metabolism, ultimately producing about thirty-six additional ATP—which is a very desirable outcome from the endurance standpoint. Or, it can linger outside the mitochondrial membrane, in the muscle cell's cytosol, where it will be converted into lactic acid. You have no doubt heard about lactic acid. Its notoriety, while deserved, is somewhat overblown. Lactic acid plays the central role in determining an athlete's endurance, so we will spend some time becoming familiar with it.

When a lactic acid molecule is formed from pyruvate, it immediately dissociates into a lactate molecule and a hydrogen ion. Lactate and pyruvate molecules contain useful energy in their chemical bonds. That hydrogen ion however is an acid and its unchecked accumulation will upset the delicate acid/base balance of the cell. The conversion of pyruvate into lactate and that hydrogen ion represents a temporary dead-end to the metabolic process. Excess accumulation of these metabolites causes a reduction of glycolytic ATP production. We feel this as muscular fatigue and a slowing of our pace.

The fate of pyruvate then is a crucial determinant of the endurance performance of an athlete. If pyruvate takes the aerobic fork in the metabolic road and enters the mitochondria, it can produce a bunch of valuable energy to further sustain physical activity. If it takes the dead-end fork in the road, ATP production will eventually slow (eventually halting) and so will the athlete. This is one mechanism of fatigue that all endurance athletes are familiar with.

WHAT DETERMINES THE FATE OF PYRUVATE?

At lower intensities, pyruvate production is small enough that most of it can be taken into the aerobic pathway. In this case the athlete can be said to be operating below his or her aerobic capacity. In other words, there is enough mitochondrial capacity to take up and utilize the pyruvate production, thus preventing its accumulation and subsequent conversion to lactate. In these conditions, lactate production is minimal. Importantly for the endurance athlete, the aerobic metabolism of fat produces no lactate, and its potential metabolic downside. This makes fat the more desirable fuel in endurance sports.

To support higher intensity (speed) ATP turnover must increase. This increase will be come from the breakdown of fats and, increasingly, of sugars. As speed

Lactate as a Problem

Limited Slow Twitch Equals Aerobic Deficiency

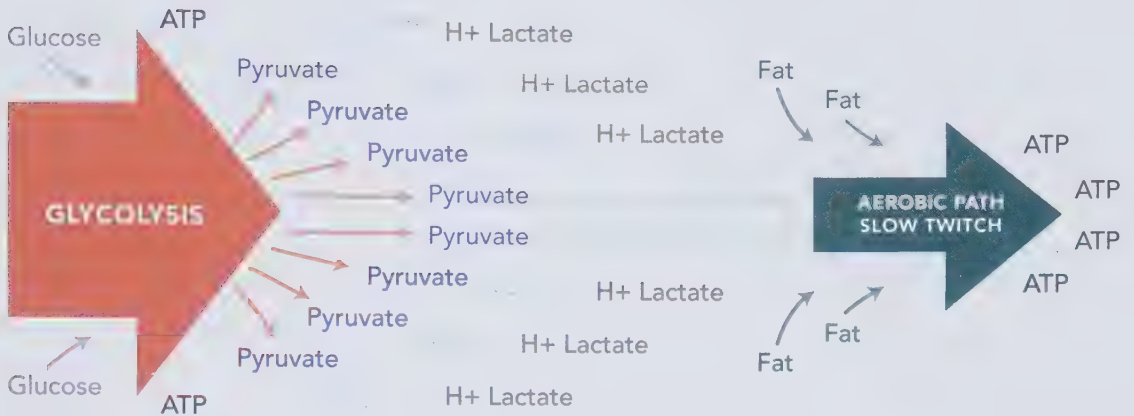


Fig. 2.4 Lactate as a Problem

A metabolic traffic jam occurs when an athlete has insufficient aerobic capacity to shuttle all the pyruvate being produced by the anaerobic metabolism back into the mitochondria where it can contribute to ATP output. This pyruvate back-up converts to lactate and a hydrogen ion and increases the acidity of the cell thereby slowing glycolysis. We perceive this disruption of cellular homeostasis as fatigue—our pace slows and our effort increases.

Lactate as a Benefit

Abundant Slow Twitch Equals High Aerobic Capacity

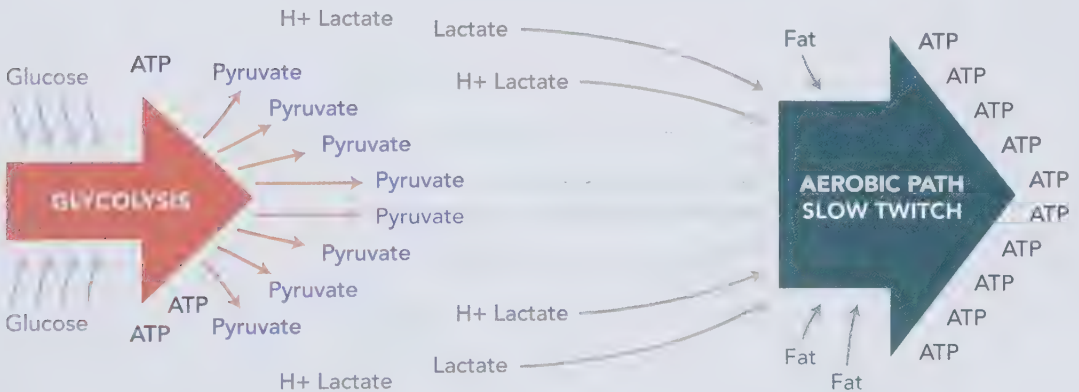


Fig. 2.5 Lactate as a Benefit

An athlete with a high aerobic capacity has the ability to utilize this pyruvate, preventing the back-up and consequent repercussions. Cellular homeostasis can be maintained at higher energy outputs. This means you can run faster for longer before needing to slow down.

continues to increase and meeting the ATP demand relies more on glycolysis, the capacity of the slow twitch muscle mitochondria to take up pyruvate will eventually be overwhelmed and lactate and hydrogen ions will accumulate at an ever increasing rate. The following three scenarios will each play out as you move up the intensity/speed scale from low to high.

1) Low Speed. At low intensities, predominately using slow twitch (ST) muscle fibers, the rate of ATP production is modest. Much of the ATP will come from the breakdown of fats, which does not produce any lactate. ST fibers have a high concentration of mitochondria and are able to take up the pyruvate from glycolysis for the aerobic metabolism. Any lactate produced can be used in the heart, liver, and adjacent active ST muscles. At this speed, you are operating below your aerobic threshold (AeT). (See note 1 below.) This intensity range represents Zones 1 and 2 in the conventional five-zone system. Zones will be explained in more detail in the Methodology Section on page 84.

2) Medium Speed. As more speed is called for, you begin to recruit some fast-twitch (FT) muscle fibers. FT fibers contract with more force, so are used when extra force is needed. They are less fatigue resistant than ST muscles because they are not as well endowed with mitochondria. They rely largely on glycolysis for their energy needs, producing more pyruvate. So, pyruvate and lactate begin to accumulate. As long as that accumulation is less than its removal (by the ST muscles, heart, and liver), this speed can be maintained. This is a comfortably hard pace sustainable for many minutes up to one hour. This intensity range equates to Zone 3.

3) High Speed. As ATP turnover continues to increase, you eventually pass a critical point called the Lactate Threshold (LT). (See note 2 below.) Beyond this intensity, the rate of pyruvate/lactate production exceeds its removal rate. This lactate accumulation—plus those pesky hydrogen ions increasing the acidity of the muscle—plays havoc with the happy condition known as homeostasis in the muscle cells. This leads to a slowing of glycolytic ATP production required by those very same FT fibers that are allowing the high speed. The drop in ATP output means the pace must slow after a few minutes of operating above the LT. If not by consciously slowing, the pace will involuntarily slow. Every endurance athlete has experienced this form of fatigue. This intensity is covered by Zones 4 and 5.

The three numbered paragraphs above corresponding to different metabolic states inside the working muscles should form the basis for every zone system used for grading intensities. We will explain a metabolically based zone system later in the book.

Lactate as a Measure of Intensity

Since lactate accumulates in lockstep with pyruvate, it can serve as a marker of intensity. More lactate hanging around in the blood indicates a higher production of pyruvate by the glycolytic metabolism. Coaches and sports scientists have long

used lactate concentrations in the blood as a measure of intensity. The subject of lactate measurements can, and does, fill a book (*The Science of Winning* by Jan Olbrecht).

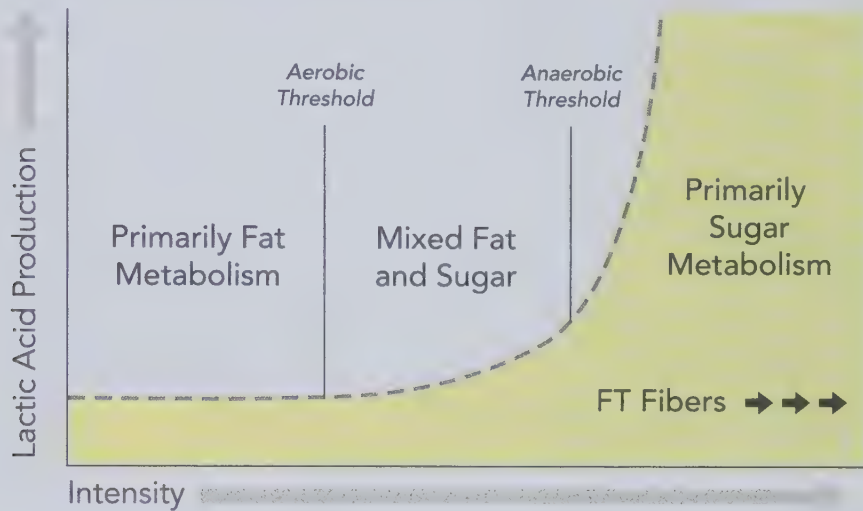


Fig. 2.6 Intensity Versus Lactate

Increasing exercise intensity is fueled by increasing sugar metabolism (glycolysis) and a corresponding drop in the percentage of fat used for fuel (lipolysis). This increased glycolysis means an increase in lactate production. By measuring blood lactate levels, one can get a good idea of where an athlete is operating along this metabolic spectrum.

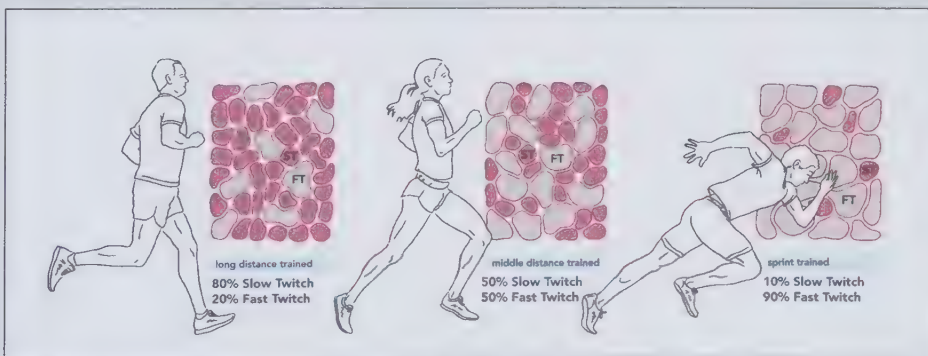


Fig. 2.7 ST and FTa, and FTb Muscle Fibers

Shown above is the relative amounts of slow (red) and fast (pink) twitch muscle fibers in the leg muscles of different running populations. While genetics play a significant role, the type of training one engages in year after year also seems to influence this mix.



Measuring blood lactate after a climb of Mount Bachelor in Oregon. Photo: Steve House Collection.

Note 1

The aerobic capacity is indicated by the athlete's AeT. This is sometimes also referred to as Onset of Blood Lactate Accumulation because it is the intensity of work at which blood lactate concentration rises 1mMol/L above baseline levels. This point is best determined with a laboratory test. We'll cover these testing procedures in the Methodology Section on page 152.

Note 2

Lactic Threshold, Anaerobic Threshold, Maximum Lactate Steady State, and Functional Threshold are all names for essentially the same metabolic event. Just remember that this is where lactate production exceeds its rate of removal. Beyond this point, exercise can be sustained only for a few minutes at a time before fatigue will cause slowing.

Summary of Metabolism as It Relates to Endurance

Before moving on to the implications of what you've just read, let's take a moment to restate what we know so far:

- Your speed is a function of your muscle's ATP recycling rate.
- ATP can be produced via aerobic and anaerobic metabolic processes.
- Anaerobic and aerobic ATP output combine to meet energy demands.

- The aerobic pathway is mainly used at lower-intensity, long-duration exercise.
- The anaerobic pathway should* be mainly used for higher intensities.
- The anaerobic ATP production is limited in duration by the accumulation of pyruvate/lactate.
- *ST muscle takes up pyruvate and uses it for fuel in aerobic metabolism.* We'll come back to this very important point.

**Should, because as we will soon see, many developing athletes must rely on glycolysis even at very low intensities. We will address the topic of Aerobic Deficiency Syndrome in a few pages.*

The Implications for Performance

From a performance standpoint, the metabolic fate of that pyruvate molecule is the essential event in determining endurance in events longer than two minutes. Physiologically speaking—the end goal of most endurance training should be to encourage that pyruvate molecule to enter the mitochondria and undergo aerobic metabolism. How that is achieved is what we are going to spend most of the rest of this chapter and much of this book looking at.

We have talked at length about the pyruvate molecule and this may be confusing to some. “Isn’t lactic acid the demon of performance?” Lactic acid gets the bad rap, but it’s worth remembering that *lactic acid only forms when pyruvate hangs around and does not enter the aerobic metabolism*. So, elevated pyruvate means elevated lactate. With proper training, you can increase the speed/intensity at which pyruvate production exceeds its removal into the mitochondria, causing that accumulation of lactate. This point is defined as the Lactate Threshold (LT). Elevating the speed at the LT results in improved endurance performance.

We have previously defined this intensity as the LT. There are two ways we can raise the LT:

1. Reduce the production of pyruvate/lactate in the first place.
2. Increase the rate of removal of pyruvate/lactate from the working muscles.

Each of these requires different and to some extent mutually exclusive forms of training. This is the reason that understanding something about intensity and its effect on the metabolic systems with regard to endurance is important if you hope to maximize your performance.

Step 1: Reduce the Production of Lactate

Goal number one for any endurance athlete must be to maximize his or her aerobic capacity. Aerobic capacity is a term used to describe the limit of the aerobic

metabolism to produce ATP. It is largely a function of the adaptations in muscle cells and is enhanced through training. Athletes with a high aerobic capacity can maintain relatively high speeds over long distances with a low metabolic cost since they create much of the needed ATP using the aerobic pathway, minimizing lactate accumulation. This system of training was described in detail by Soviet sport scientist Dr. Yuri Verkhoshansky in his article “Block Training System for Middle Distance Runners” in the mid-1980s where he coined the term “antiglycolytic” principle of training.

AEROBIC DEFICIENCY SYNDROME (ADS)

When you exceed the capacity of the aerobic system’s production of ATP, you will be forced to rely upon the anaerobic system to make up the deficit. As you now know, that reliance will be self-limiting and you will have to slow down sooner or later. Athletes who overuse high-intensity training methods will usually have a low AeT because their aerobic metabolism has become detrained while the anaerobic metabolic pathway has become very powerful. We call this condition Aerobic Deficiency Syndrome (ADS). It is reflected in a slow (often shockingly slow) AeT pace. These athletes will be dismayed that despite all their hard training, they still can barely get above a jog or even a walk before exceeding their AeT. These athletes typically respond with disbelief: How can they possibly be getting any training benefit from such a slow (perhaps even walking) pace? This low aerobic capacity has only one cure: a high volume of low- to moderate-intensity training. To determine if you have ADS, read “The Ten Percent Test” sidebar on page 91.

To give credit where credit is due: ADS was first described to the author Scott Johnston in 1985 by the coiner of the term, Dr. Phil Maffetone, in explaining a common phenomenon he saw frequently among runners. We have used it since to describe the condition of underdeveloped aerobic conditioning. Maffetone is perhaps most famous as the coach of the legendary Ironman Triathlete Mark Allen. He places a very high emphasis on rectifying ADS and maximizing this aerobic capacity during the crucial aerobic base-building period before the athlete engages in high-intensity training. He calls this period the Patience Phase, with good reason. Aerobic adaptations are slow to accumulate, and patience is needed. Approach this training with diligence rather than ferocity and patience will be rewarded.

This condition is more common than you might think, and we’ll examine one athlete’s case.

ANTIGLYCOLYTIC TRAINING METHOD

Duration is the single biggest stimulus from any exercise. The more you train a certain way (within your ability to absorb the work), the more adaptation you

A Real-World Example of ADS

We often see athletes who have ADS. Some of them have even managed to achieve fairly high levels of performance despite this handicap. For successful athletes it can be a difficult pill to swallow that they need to take what appears to be a step backward in order to move forward.

A few years ago a top US Cross Country skier approached Scott Johnston for help with his training. Despite his very impressive VO_2 max test numbers (90 ml/kg/min), this skier was suffering from ADS due to the type of training he had been using. His performance progression had stagnated for two years, which is typical for athletes relying too heavily on medium- and high-intensity (Z3–Z4) training. The adjustment to his training was drastic and meant taking two steps backward before he could advance. The hardest thing about this training prescription was the embarrassment he felt by training so slowly.

With the new training Johnston prescribed, during his four-month base period the skier managed to drop his AeT pace from 8:30/mile (5:16/km) to 6:15/mile (3:53/km) with a corresponding heart rate increase from 135 to 158. This occurred as a result of doing *all* his aerobic training under his AeT. His speed was no longer embarrassingly slow and represented a dramatic improvement. He was a fast responder due to the high volume of this low-intensity training he was able to do. He had gotten his aerobic house in order. The implications of an improved AeT for him were:

- His basic aerobic training pace moved significantly closer to his racing pace, so even his easy base training had a more race-specific training effect. This improved mechanical economy at race pace because he was spending more time near to race pace.
 - Because of the higher speed that he could now sustain, the neuromuscular stress of even Z2 workouts went up, thus requiring more recovery training between these AeT sessions. In the beginning of this program he could do six to seven Z2 workouts a week; by the end, that had to drop to one or two per week because the pace was now *fast*.
- Once this aerobic foundation was created, the next year we added more race-specific training. He greatly improved his personal bests and was selected for the 2014 Olympic team.
- The lesson is that, if you suffer from ADS, then the AeT pace is going to be very slow and very far from race pace. Accept that this is all the power your aerobic system is capable of in its current deficient state. No ifs, ands, or buts about it. Faking it by running at a faster pace to soothe your ego will only slow the development of that aerobic engine.
- He could produce a lot more power/speed aerobically than he could when he started this program. His metabolic efficiency (fat adaptation) was greatly improved. His body experienced less metabolic stress and more energy per gram of fuel.



Michele Maccabelli finding long runs on the ridges of Monte Baldo, above Malcesine, Lake Garda, Italy. Photo: Martina Valmassoi

will cause. One of the strongest training stimuli for increasing the aerobic recycling rate of ATP is glycogen depletion in the ST muscles. This is best done with long-duration, low- to moderate-intensity training sessions because it takes long durations to deplete the glycogen of these already well-endurance-adapted ST muscle fibers. Short, high-intensity training depletes the glycogen in the less well aerobically adapted FT fibers but is too short in duration to have much training effect on the important ST fibers. This is why endurance training has traditionally contained so much relatively easy intensity training volume.

If you want to reduce the production of lactate, both by using more fat for fuel and by being able to produce more power from the aerobic system, you need to do a lot of long-duration, low-intensity work. This is the reason the common classic training method across all endurance sports is called building an aerobic base. An athlete with low aerobic capacity (small aerobic base) can never fully maximize his or her endurance performance no matter how much high-intensity training is done.

Step 2: Increase the Rate of Lactate Removal from the Working Muscles

When the ability to run or ski fast without producing much lactate is maximized we can focus on lactate removal improving endurance.

REMOVING LACTATE: THE LACTATE SHUTTLE

From an endurance performance standpoint, being able to make use of pyruvate/lactate in muscles is one of the most important training adaptations. Research done by Dr. George Brooks of the University of California, Berkeley, described a mechanism he termed the “lactate shuttle,” which remains the best explanation for determining the LT. He found several proteins that are responsible for intra- and inter-cellular lactate transport. These transporters shuttle the lactate from where it is accumulating and causing problems to the mitochondria in adjacent ST muscles, where it can be utilized as fuel.

Increased levels of lactate in the muscles acts to signal an increase in these transporter proteins. So, high-intensity training that causes lactate accumulation enhances the lactate-shuttle process.

This mechanism has gone a long way to explaining what endurance coaches have noticed for decades: When a large volume of low-to-moderate-intensity (aerobic) work is judiciously supplemented with higher-intensity training, that improves the shuttle process, and hence the highest LT and the best endurance results are obtained. *The key point of misunderstanding for many athletes and coaches is how much of the high-intensity training to apply.* It turns out to be much less than is commonly assumed and we'll come back to this soon.

When steps 1 and 2 are combined, you can see how aerobic metabolism acts like a vacuum cleaner sucking up all that pyruvate and thus reducing the accumulation of lactate. Slow-twitch muscles are the ones with the biggest vacuum cleaners, so they are responsible for the bulk of the lactate removal in the working muscles. The greater the ST fibers' aerobic capacity, the bigger the vacuum cleaner.

Enhanced Lactate Removal and Endurance: The Metabolic Key to Improving Endurance Performance

Given the above discussion, the best method for improving endurance is to do these two things:

1. Increase the aerobic capacity (make a bigger vacuum cleaner) so that you have a larger reservoir of mitochondria in the ST muscles to take up and use excess pyruvate and lactate to produce useful energy.

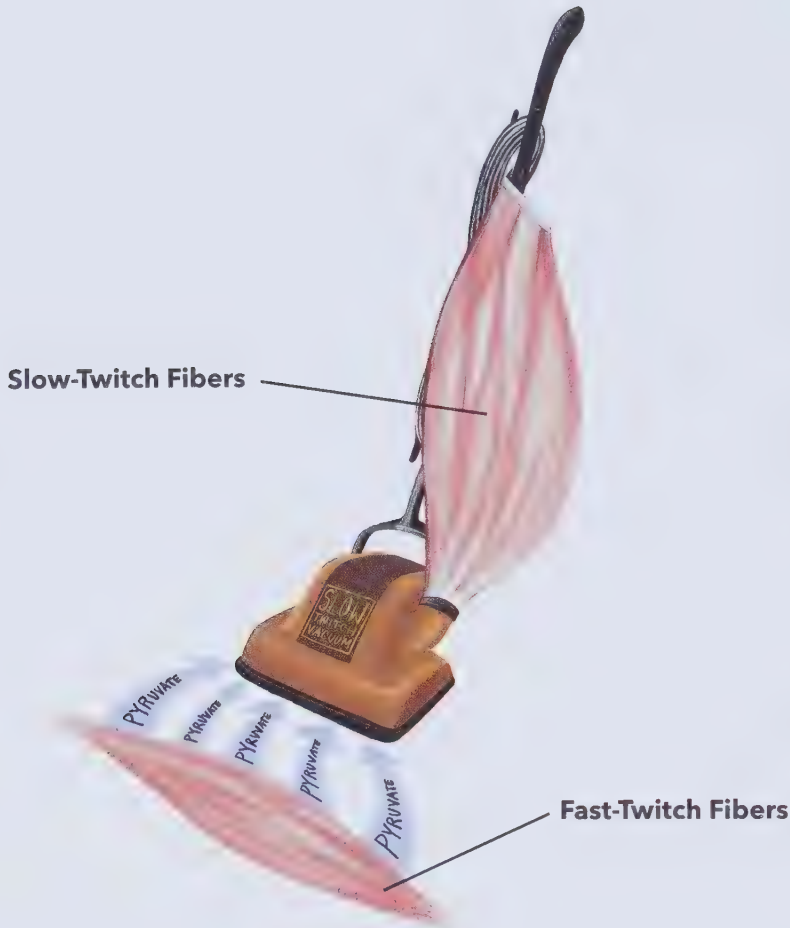


Fig. 2.8 A Bigger Vacuum

The aerobic metabolism, especially in the slow twitch muscles, acts like a vacuum cleaner, sucking up all the “bad” by-products of anaerobic metabolism. The bigger the capacity of this vacuum cleaner (aerobic capacity), the more anaerobic metabolism can contribute to the overall energy production before you slow down.

2. Improve the lactate-shuttle mechanism (make the vacuum more powerful) so lactate that is produced at higher intensities can be effectively converted into useful energy rather than gum up the ATP works.

Achieving this is tricky because doing the first requires one type of training while doing the second requires a very different kind. Increasing aerobic capacity requires a high volume of low-to-moderate-intensity work. Improving the lactate shuttle, on the other hand, requires high-intensity training that produces significant levels of lactate.

The bigger and stronger the vacuum cleaner, the more high-intensity work that can be done, and the more power/speed an athlete can produce for longer.

ATHLETE STORIES

Nickels and Dimes

Rod Bien

I have run eighteen 100-mile races at this point, and I have thrown up during every single one. Not just once per race, but ten to twenty times per race. When my body becomes stressed, my GI system is the first to shut down. I have tried numerous diet and fueling techniques, but all end with the same result: vomiting. Instead of dropping from races—or quitting the sport altogether—I’ve come to accept this physiological quirk of mine. In fact, I’m so used to it now that sometimes I’ll throw up right in stride.

I’ve been running ultramarathons for eighteen years, as well as numerous adventure runs. In all of these events, but especially during mountainous 100-milers, I have experienced physical and mental difficulties—including the now-familiar indigestion. But my desire to keep going always overrides any discomfort. I’ve had to DNF (did not finish) only once, and that was because I felt the outcome could be dangerous.

I’ve noticed that when runners DNF, their reasons for quitting are often very specific and numerous, with the excuses accumulating from the very beginning. The pity party may start innocently enough: “I had the wrong breakfast.” Then a couple miles in: “The trails are muddier than expected and I’m slipping around.” At mile 15: “I have a hot spot on my heel.” At mile 30: “My crew didn’t meet me and I didn’t get my normal gels.” As the race or adventure goes on and mental strength diminishes further, all of these little items start to “nickel and dime” a runner toward a DNF.

One of the intangibles that separates better endurance athletes from the rest is how we handle these challenges—small to serious—that almost everyone will experience at some point during a difficult race or climb. Around mile 35 during the 2011 San Diego 100, my stomach rebelled, and I dropped from second place to fourth or fifth. Demoralized, I started consulting the “rolodex of excuses” for why I should quit. But instead of settling on one, I decided to problem-solve my way out

of the situation. I nibbled on some avocado. I was still sick, but I concentrated on the fact that I was running through some beautiful mountains and just needed to get to the next aid station, which was five or so miles away. Once I got there, I felt a touch better and focused on the goal of pressing on until the sun started to fade. Once the shadows lengthened, my spirits and stomach rebounded. I ended up finishing in second place with a great race to look back on. Driving away those negative thoughts and concentrating on short-term goals and positive reinforcement got me through.

Part of being a successful endurance athlete is recognizing and trying to solve challenges without letting them stack up against you mentally and physically. It all comes down to outlook: Do you let the trivial issues overwhelm you, or do you shrug them off and keep going? Do you DNF due to stomach issues, or do you throw up without breaking stride?

Going into an endurance event, you need to have a positive attitude and excitement about the adventure—but it is also extremely important to understand that there will be many physical and mental obstacles you will have to overcome along the way. While dealing with them, keep going and do not allow yourself to sink into a dark mental place. Don’t nickel and dime yourself out of the satisfaction of crossing the finish line.

By eliminating the negative voices in your head and concentrating instead on what you can control, you’ll develop the mental tenacity to become a stronger, and ultimately faster, endurance athlete. ■

Rod Bien is a Patagonia Ambassador and one of the top masters ultrarunners in the country. He has been racing ultramarathons since 2001 and has won races and set course records at 50Ks, 50-milers, and 100-mile endurance races. He is a husband, a business owner, and a very proud parent of three kids.

The greater the aerobic capacity of your ST muscles, the more lactate that can be shuttled into those muscles' mitochondria and turned into useful energy. This allows you to sustain higher outputs for longer. You have no doubt heard the various terms of adulation heaped on amazing endurance athletes: "He has a great aerobic base" or "She has a really big motor." Now you understand the physiology behind those comments. Here is the counterintuitive rub: *To improve your high-intensity endurance you must maximize the aerobic capacity of those unsexy, low-intensity, low-power, slow-twitch fibers.*

THE ALLURE OF HIGH-INTENSITY TRAINING: THE COUNTERINTUITIVENESS OF TRAINING FOR ENDURANCE

The type of training determines the type of adaptations that are induced. The most important adaptations to lower-intensity aerobic training are structural, meaning that you grow new protein structures in your body: denser capillary beds in the trained muscles, increased mitochondrial mass, and changes to the heart. These are slow to occur and take from weeks to months to show significant changes but can continue for years, which is one reason that endurance athletes often have very long careers, continuing to improve even into their forties.

Adaptations to higher-intensity training that enhance the anaerobic metabolic output are of a functional nature. These are changes to the function of pre-existing structures. They are primarily changes in glycolytic enzyme concentrations or to the lactate transporter proteins. These adaptations occur within hours of a workout and can show significant improvements within a few days and weeks.

Due to the rapid response to high-intensity training stimuli, adaptations to this sort of training make themselves felt very quickly. Especially as compared to the weeks and months of patience required before seeing significant gains in aerobic capacity. The positive reinforcement effect of seeing rapid gains leads many athletes and coaches to overuse high-intensity training. This stuff feels like training should feel ... hard and tiring. It gives tangible and rewarding results that can be quickly seen and felt. What's not to like about that?

Training dominated by high intensity often fails in the long term to produce the desired results, not so much because the athlete is doing too much high intensity but because that person is doing too little low intensity. The final result of this strategy, usually seen after just a few very exciting and rewarding weeks of high-intensity workouts yielding big gains, is first a plateau then a reduction in performance. This happens because those lowly, low-power ST fibers are not getting enough training stimulus and so their aerobic capacity begins to drop off. The vacuum cleaner gets smaller. Even though the aerobic capacity plays a supporting role to the anaerobic star of the show, those ST fibers' aerobic capacity role is critical in setting the endurance limit.



Sabina Parigian rips down the trails of Longs Peak, Rocky Mountain National Park, Colorado.
Photo: Fredrik Marmsater

This performance plateau or drop is often interpreted as a need to add more intensity to the program when in fact, just the opposite is needed. The aerobic base has eroded and endurance will not improve until that base is restored. Maintaining sufficient aerobic-capacity workouts during the high-intensity phase of the training program is essential to long-term performance gains.

The reason for this confusion is that our intuition is wrong. When an athlete nears the limits of his or her endurance, it is at higher intensity that the limits become apparent. When you can't sustain the pace as the hill becomes steeper or you are unable to make your legs respond to the challenge of outkicking your rival in the last minutes of the race it is quite intuitive to assume that the endurance limitation is caused by some deficiency in your training at this race-specific intensity. Intuition tells us that if we are weak at something, we need to train it more.

But not if the vacuum cleaner is too small.

The Big Motor

From the above you can see that both low- and high-intensity training are important to proper endurance training. Neither alone will suffice to maximize performance. It is the balance and timing of the application of these methods that is so critical to this end goal.

Increasing aerobic capacity has major benefits to all athletes regardless of the duration of the event they are training for.

- Most interesting to endurance athletes is that increasing the size of the aerobic vacuum as defined by the AeT will have very direct and profound effects on increasing the athlete's endurance or LT (the power of the vacuum).
- You will also see improved recovery rates from workouts, including very high-intensity ones. As you will read in Section Two, this is one of Kílian's biggest advantages over most other athletes. Having a big aerobic motor will allow you to handle more anaerobic work. Since all training adaptations occur in direct relation to the amount of time spent doing the training, this means you can do even more anaerobic work as a result of improving your aerobic system.

The whole fitness house needs to be constructed from the ground up on a solid foundation (high AeT) before worrying too much about the upper floors.

The Other Two Legs of the Endurance Tripod: Economy and VO₂ Max

We have devoted a lot of ink on the previous pages to the role of metabolism in endurance. It may seem like we are giving short shrift to movement economy and VO₂ max, but we will address them now in greater detail.

MOVEMENT ECONOMY: THE VALUE OF GOOD TECHNIQUE

The mechanical component of economy, often referred to as technique, varies from sport to sport and is best learned from visual cues and direct coaching, not through a book.

Technique in anything, from playing the piano to running fast on rough terrain, is something that is acquired only through many, many hours of specific practice. We learn through repetition, creating motor nerve firing patterns that result in the desired movements. By patterns we mean that many thousands of motor units (nerve/muscle combinations) must fire in exactly the right sequence with the right magnitudes to create these skills that we are seeking.

If your training is predominated by running on flat surfaces in urban areas, your nervous system gets hardwired for these types of movement patterns. The motor cortex of your brain has developed a library of what are called engrams to choose from for these firing patterns. It is quite good and efficient and will automatically select the appropriate engram without any conscious effort. But when you switch to rough terrain or downhills or skimo, the library shelves are nearly empty, and you are forced to use the conscious control of your frontal cortex for running. This conscious control is slower, less efficient, and tiring.

Simply stated, you have to run a lot of downhills and rough terrain to become proficient at them. You will not learn to play Chopin by practicing “Chopsticks.” You’ll have to expand your engram library, which is challenging and time consuming. But there is no other way to do this. For a discussion of metabolic economy see the next section on fat adaptation.

VO₂ MAX

The VO₂ max is the least trainable of the three endurance components for the well trained. Of the three, it has the poorest correlation to performance. The preponderance of evidence indicates that, to a large extent, it is genetically predetermined. Since you can’t choose your parents, it seems rather futile to become overly concerned with this number. We have coached athletes competing at the highest levels of international competition who have had exceptionally high VO₂ max values and turned in rather modest results. At the same time, in the same races, we have coached athletes who had rather modest VO₂ max values who achieved exceptional results. This extensive firsthand experience has made us skeptical of the importance placed on a single number as a predictor of performance. More importantly, it should cast doubt on the common practice of training to improve VO₂ max (rather than performance). There is a lot written about the subject, and we provide the sidebar on the next page for those who seek additional info.

Fat Adaptation, Metabolic Economy and Endurance

Coaches of endurance athletes, across a broad spectrum of sports, have known for many years that well-trained athletes can utilize fats better and at higher (even maximum) intensities than studies indicated was theoretically possible. It was assumed by coaches that this adaptation was an important key to success in endurance sports. But research didn’t support the coaches’ experiences. Several new scientific studies now substantiate the coaches’ beliefs. (See note 3 for studies.)

How the Scientists Got It Wrong

For decades, exercise scientists have told us that fat played a significant role in total energy use only at low-to-moderate-exercise intensities and that fat’s contribution was theoretically capped at a paltry amount. This meant that energy demand for high-intensity exercise could be met only with carbohydrates. They were partly correct: this is what happens in the general population and, to a lesser extent, in the recreationally trained athletes who were the subject of their tests.

Misunderstanding VO₂ Max

With the way that the term VO₂ max gets tossed around in both the popular media and in scientific studies on endurance, you could be forgiven for thinking that this single number was a direct proxy for endurance performance. It might then come as a surprise to learn that VO₂ max is not a measure of endurance at all.

Measuring the maximum amount of oxygen that the body could take up and utilize was first carried out by Archibald Hill, PhD, in the early twentieth century. Both the measure itself and the theory that he proposed for the limits it places on human performance won him the Nobel Prize in 1922 and have formed the basis for a huge portion of the study of exercise for the past one hundred years. However, neither Hill's theory nor the numerical quantity were intended to describe endurance. It is only recently that the popular press, and by extension the public, came to see VO₂ max as somehow identical to, and indicative of, endurance performance. Despite its popularity as a metric, this single number, while it does relate to endurance potential, does not define aerobic endurance.

The term VO₂ max defines the upper limit of an individual's aerobic power. It is analogous to maximum strength in some important ways. Just as you can't exert your maximum strength many times in succession, VO₂ max can't be sustained for long. But, just as having a high maximum strength would give you a greater potential of lifting a lighter weight many times in succession (muscular endurance), having a high VO₂ max also allows for the potential to develop a high level of aerobic endurance.

Research suggests that the most important factor in determining VO₂ max is the amount of oxygen-carrying blood ejected with each heartbeat, called stroke volume.

Long-term studies, along with personal testing by author Johnston, have demonstrated that elite endurance athletes will often show no change (and even a reduction) in VO₂ max during a competition season and over their careers while at the same time seeing marked improvements in performance. These gains in performance come from improved economy and speed at the LT (the other two legs of the endurance tripod).

VO₂ max is what is called a "first wave response" to training. This term refers to a quality that tends to develop quickly in the young and those new to endurance training. That is largely because the heart responds rapidly to training and increases the aforementioned stroke volume and other characteristics that increase the volume of blood ejected from the heart.

Many scientific studies aimed at endurance will measure changes in VO₂ max in test subjects after several weeks of certain training protocols. The VO₂ max has become the standard metric for measuring cardiovascular fitness because it is easily measured and can be compared across the test subjects as well as across populations—not because it is the best predictor of endurance. The unintended consequence of this has been to make VO₂ max synonymous with endurance in the minds of the lay public. Other factors that have contributed to this misunderstanding are:

- Most studies are relatively short term because it is hard to find subjects willing to engage in invasive studies lasting for months or years. So most tests use training interventions that can show results in a short time. Functional adaptations from high-intensity training occur quickly. The result is often an increase in VO₂ max across the study subjects.

- Most studies use recreationally active or untrained individuals since it is hard to find high-level athletes willing to forego their normal training to engage in a study. Almost any training will increase VO_2 max in an untrained population.
- The results of these tests are presented as average changes in VO_2 max. Some subjects may have seen a decrease but as long as the average is significantly positive, the test can be held up as proof that such and such a training protocol works. The all-important individual nature of training response is not taken into consideration in these studies.

All these factors lead to an overemphasizing of VO_2 max's importance in the public's mind. Many of these studies compare some high-intensity interval training (HIIT) program against lower-intensity, continuous training protocols. This oversimplification and misunderstanding has led to a rash of training schemes promising to raise VO_2 max. Since HIIT affects systems that respond quickly and dramatically (functional adaptations), the HIIT protocols almost always come out looking like a shortcut to fitness. Mix this misunderstanding of these studies with a media bent on discovering the next new thing and a time-crunched public looking for shortcuts, and you have the perfect storm of the HIIT fitness craze. These programs, often gym-based, are social and great for getting people exercising. However, there are no world-class endurance athletes in the major sports of running, rowing, cross-country skiing, or skimo using an exclusive HIIT training program. Enough said?

High-intensity training, while important for any endurance athlete, should be

used as a supplement to, and not a substitute for, aerobic base training.

The takeaways from this are important:

- 1) Beware whenever you read that some study "proves" that such and such HIIT methodology is "the" recipe for successful training because it raised the VO_2 max of its subjects by X percent.
- 2) Don't believe that there is a newly discovered shortcut to endurance fitness that only these overhyped fitness fads have discovered. If that were true, professional endurance athletes of all stripes would be beating down the doors of their local CrossFit.
- 3) The history of endurance training is the story of trial and error over hundreds of years by thousands of coaches and millions of athletes. While the evidence is anecdotal, the population size is so big that coaches can weed out what does and doesn't work. New stuff is tried all the time. Good ideas stay, bad ones get rejected or modified until they are good ideas. These ideas must stand the toughest test of all, the stopwatch, in the highest levels of competition.

Note:

Bassett DR Jr, Howley ET, "Limiting factors for maximum oxygen uptake and determinants of endurance performance," *Medicine and Science in Sports and Exercise* 32, no. 1 (Jan. 2000): 70-84.





This scientifically supported understanding became enshrined in sports science literature, and you will still find nutrition and exercise scientists today who are unequivocal in stating that once the athlete reaches his or her LT, fat metabolism will have ceased to contribute any energy to the overall exercise demands.

New studies (see notes below) have shown that the rate of fat usage is much higher (up to twice as high) in well-trained athletes and can be sustained to maximum intensities, contrary to the conventional wisdom. To put this correction in perspective, that's like increasing your car's horsepower and its fuel economy at the same time!

Becoming Fat Adapted

Becoming fat-adapted is one of the simplest to implement training prescriptions you will read in this book, and the returns from it will be significant for the effort involved. Recall from earlier that glycogen depletion is one of the most powerful training stimuli responsible for aerobic adaptations. Glycogen is depleted from the all-important ST muscles during long training sessions. Engaging in these long-duration, low-intensity workouts also enhances fat adaption. You can boost this adaptation by doing some of your aerobic capacity-building workouts in a fasted state. If you train in the morning before consuming any calories, you will be in a mildly glycogen-depleted state. This will force your muscles to make better use of the available fat for fuel and increase their aerobic capacity. Consuming a diet reduced in carbohydrate calories and increased in fat will also accelerate fat adaptation. Athletes training more than twelve hours per week will see less effect from modifying their diet because the low-intensity training volume is providing ample stimulus for fat adaptation.

Two advantages are gained with improved fat adaptation: The first relates to fuel depletion. Fat stores, even in a lean, well-trained athlete, are virtually limitless (up to 100,000 calories of intramuscular fat). On the other hand, glycogen stores will rarely reach 2,000 calories even in the well trained. Glycogen is a critical nutrient even for the fat adapted. So much so that when you deplete your glycogen stores (bonk) the effect on performance is a dramatic and swift drop in power. Sipping from your limited glycogen reserves while guzzling from the bottomless fat gas tank enables higher outputs for a longer time before the low fuel light comes on.

The second and potentially greater performance benefit is reduced lactate production at submaximal efforts. As you know by now, the unchecked accumulation of lactate in the muscles leads to a type of short-term fatigue that most of us are

Previous page: Pierra Menta 2013, day two, when the race was routed across the wild and rarely seen Roselend side of Beaufortain. During a long, multiday stage race like the Pierra Menta, fat adaptation plays a major role in a successful finish. France. Photo: Jocelyn Chavy

How Much Fat Can Humans Use?

A Consensus View

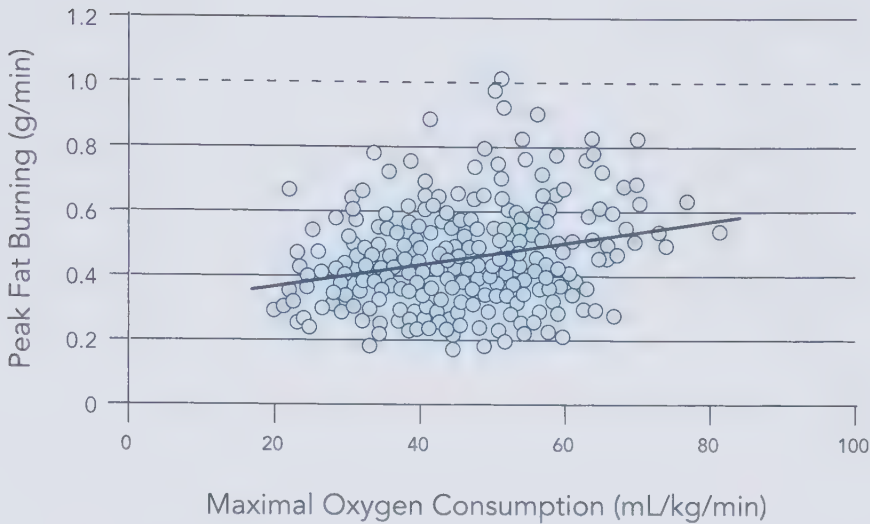


Fig. 2.9 How Scientists Got It Wrong

In the above diagram each dot represents the results of a study done on fat's contribution to metabolism at various intensities. Most of these studies were done on untrained individuals. The horizontal scale shows increasing intensity as a percentage of VO_2 max while the vertical scale represents fat use. The sloping line shows the average fat contribution in all these studies. The dashed line shows what was considered to be the maximum possible fat use (one gram/minute) which would occur at relatively modest intensity. The dogma that at higher intensities fat's contribution to overall energy production dropped to zero was handed down over decades even though coaches saw much higher rates of fat use among well-trained athletes.

painfully familiar with. You don't need to be an exercise scientist to understand that feeling—if you don't voluntarily dial back the intensity, your body will do it for you. Being able to increase your reliance on fat at higher intensity means lower lactate levels and lower perceived exertion. *Et voilà!* More economy!

Note 3:

Yeo and Carey, "Fat adaptation in well trained athletes," *Applied Physiology, Nutrition, and Metabolism* 36, no. 1 (2011).

Seiler et al., "Rethinking the role of fat oxidation: substrate utilization during high-intensity interval training in well-trained and recreationally trained runners," *British Journal of Sports Medicine* 1 (2015).

Volek et al., "Rethinking fat as a fuel for endurance exercise," *European Journal of Sport Science* 15, no. 1 (2015).

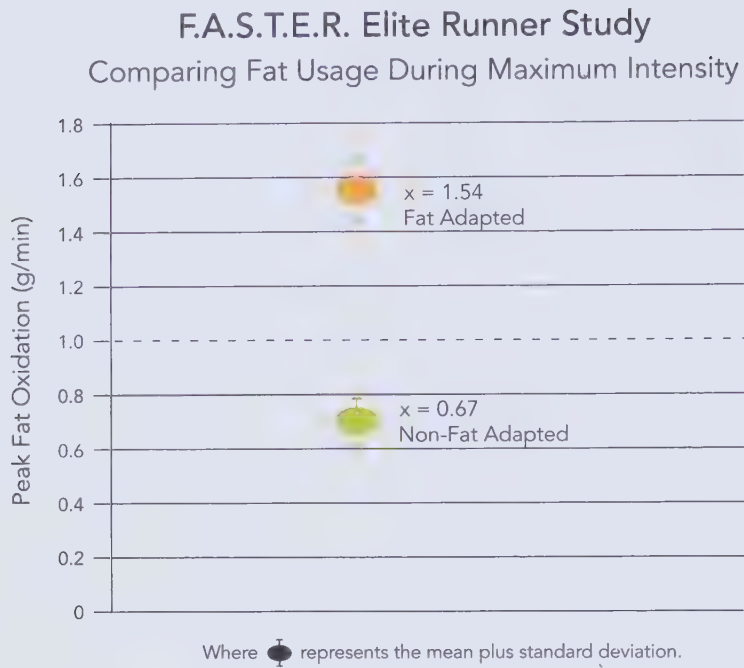


Fig. 2.10 Fat Usage During Max Intensity Exercise

Recent studies compared well-trained, fat-adapted endurance athletes (the average fat usage rate represented by the orange oval) to less-well-fat-adapted athletes (their average fat usage rate represented by the green oval). These studies show more than two times greater ability to use fat for energy by well-trained athletes and they show this fat contribution continues up to much higher intensities than was previously thought possible. Fat adaptation has positive benefits for endurance athletes. Courtesy: Voltek et al.

GRAPHICAL REPRESENTATIONS OF ALL THAT WE HAVE DISCUSSED SO FAR

The charts on the next page show the interaction of the two metabolic systems. The fuel (either carbohydrate or fat) contribution is shown as a percentage of the total energy production (vertical scale) at ever-increasing intensity (horizontal scale). Blue lines represent fat’s contribution. The green lines represent the carbohydrate contribution. The sum of the blue and green values at any given intensity will add to 100 percent. These charts show the different metabolic responses to exercise between athletes with different training states to illustrate the point that *endurance is a metabolic quality*.

In these graphs we use heart rate along the horizontal scale as a measure of intensity. The vertical scale is a measure of the percentage contribution of fat versus carbohydrate for fuel.

The comparison of these three charts shows the dramatic differences between the well- and not-so-well endurance trained in terms of fat versus carb contributions

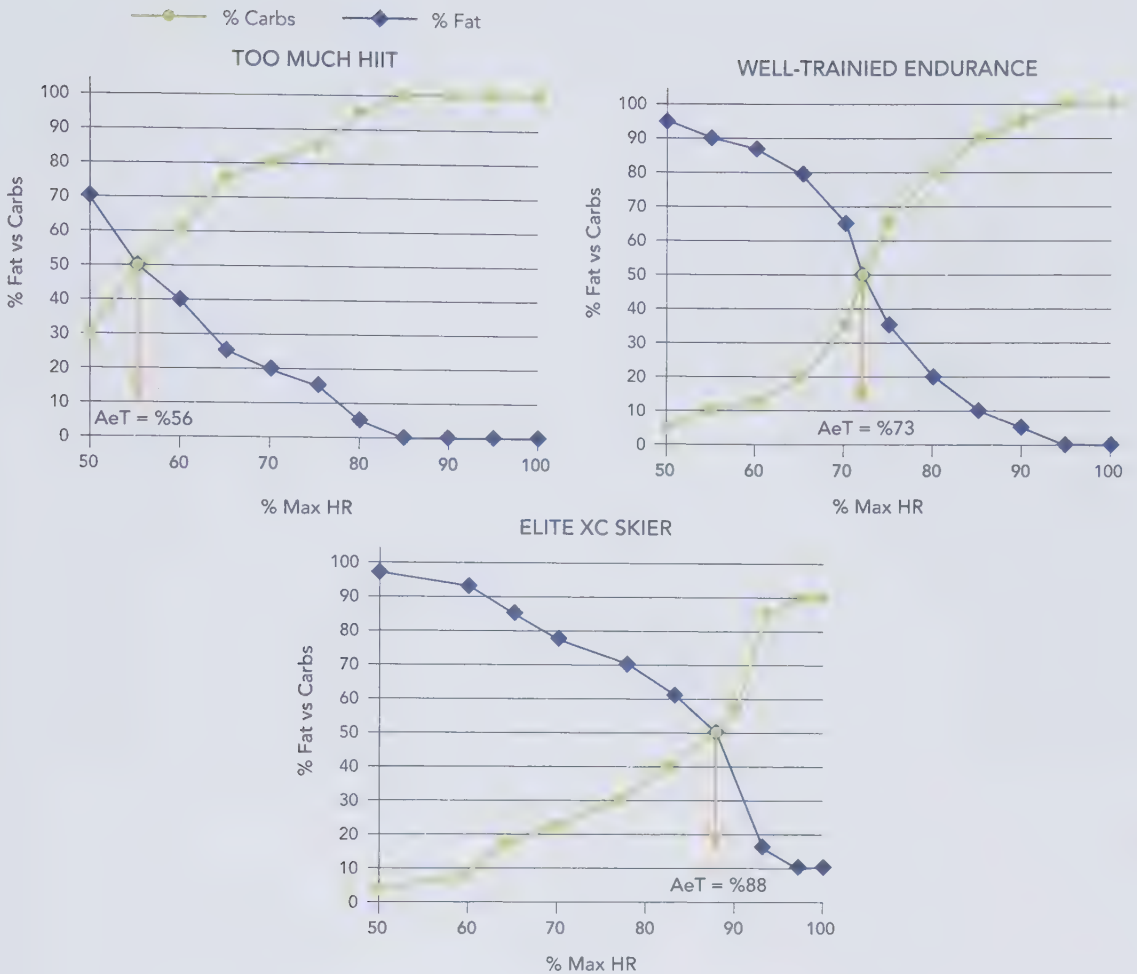


Fig. 2.11 Fat Versus Carb Burning in Athletes with Different Training Backgrounds

This figure shows the results of metabolic efficiency tests done on three different athletes: a top-ranked, cross-country skier; a recreational-level competitive runner logging 40 miles/week; and an avid HIIT adherent who did three to four high-intensity interval workouts per week with no other aerobic training. They show the relative contributions from fat and glycogen at varying intensities in a running treadmill test. The green lines indicate carbohydrate use and the blue lines shows fat use in the total energy use during the test. The horizontal axes show intensity as indicated by heart rate. The vertical axes show the relative percentage of each fat and carbohydrate. At any intensity the sum of fat and carb contribution will add to 100%. The orange arrow at the cross-over point approximately indicates the AeT for each individual. The more the aerobic threshold is shifted to the right, the bigger the athlete's vacuum cleaner and then better will be his or her endurance performance.

at different intensities. Notice that for the cross-country skier, the aerobic (fat) contribution is larger at all intensities than the other two athletes. This well-trained cross-country skier can produce much more power (ATP) from his



Janine Patitucci and Cynthia Alverson in the Italian Dolomites on a sunny summer day.
Photo: Dan Patitucci

aerobic system than the other two. The point where the blue and green lines cross usually corresponds to the AeT. The crossover point of the skier is shifted far to the right side of the chart; this indicates a very big aerobic base.

Enhancing Your Aerobic and Anaerobic Metabolic Pathways

Now that we have covered some of the basics and laid out the role metabolism plays in endurance, we are ready to talk about the principles governing the training methodologies presented in the next chapter.

By improving your ability to produce ATP faster and for longer, you can improve your endurance performance. Since the metabolic pathways are distinct, it should come as no surprise that they respond to distinctly different types of training stimuli. Understanding which stimulus to apply to achieve the desired result is the first step in understanding how to train for any event.

A strong training stimulus that is ideal for causing desirable adaptations to the aerobic system will have a negative effect on the anaerobic system. Any workout that has a powerful anaerobic training effect will diminish the aerobic system's

capacity to do work. By now you should be beginning to grasp the complexity of the systems working to supply the divergent energy needs during exercise.

The complexity of our metabolic system is also the beauty of it. We are highly adaptable. We respond to the application of different training stimuli with varied adaptations that have different effects on endurance. We can bias our metabolism toward the maximum anaerobic endurance required by events ranging from one to two minutes in length, or we can bias our metabolism toward events that last many hours and even days. But we can never optimize it to both at the same time. While this is an amazing range of potential specialization, the real lesson is that we have to *target the training we do to get the outcome we desire*.

The Uphill Athlete Training Philosophy

We follow an approach to training for endurance that has stood the test of time in all of the conventional endurance sports for more than sixty years. The following is a distillation of this successful approach:

You will never maximize your endurance potential without first maximizing your basic aerobic capacity (AeT).

In the next section we'll get deep into the various ways to do the above work.

ATHLETE STORIES

The Roots of My Athleticism

Kilian Jornet

Surrounding my house growing up were mountains and forests. I didn't see my first television until I was five years old. All the time when my younger sister and I were not in school, we would play outside together, climbing trees and jumping from rocks. It makes a big difference to be outside all the time when you are a child—to be in the outdoors, in the forest and in the snow, and get used to that kind of mountain terrain.

My mother and father were both mountain enthusiasts. My father was an alpine guide with a more traditional vision of mountaineering; he taught me technique, safety, and respect for the mountains. My mother was a longtime climber and runner, a bit more disorganized, but very experienced. She was much freer in the mountains. She liked to move fast and light.

When I was three years old, we started to do some summits as a family. We would cross-country ski up, and my parents would carry our alpine skis for the downhill. That was my introduction to ski mountaineering. By the time I was five years old, we were going up 3,000-meter peaks and practicing glacier travel with ropes, crampons, and ice axes. Around that time, I also did my first easy couloirs. In pictures from those days my helmet is so big on me that I look like a mushroom.

We backpacked across the Pyrenees—a forty-day journey—when I was ten. We did something like that every year. That's how I learned about endurance, about how to keep going over many days. By the time I was thirteen, my mother began taking me on longer outings in the mountains, and sometimes I wouldn't bring enough to eat or drink. I remember going for as many as sixteen hours without food and water, just licking water off rocks.

It was around this time that I started training for ski mountaineering. I had been doing long cycling

rides and 80-kilometer hut-to-hut runs, but without following any kind of a training plan. I entered the Sports Technification Centre in Catalonia, and my coach there sent me a plan every month telling me what I should do each day. There were two coaches, and it was maybe ten people for each coach. From the age of thirteen to seventeen, I learned about volume, intervals, strength, recovery, mountain conditions, and technique.

I was in school then, so I'd train early in the morning or after school, and then in the middle of the day I'd go to the gym or for a run. I used the weekends for longer training. Sometimes I'd bike to school, my skis in my backpack, and then after school I'd bike 60 kilometers to the snow, ski for two hours, and then bike home. Other days I'd run to and from school—25 kilometers each way. Many times my coach was upset with me because I was doing too much, even though I tried to convince her that the cycling was just my transportation to the training. I was very focused—obsessed, really.

I have always been good at recovering, so I was able to assimilate this high volume of work. I think it comes from when I was a kid, playing outdoors all day and doing long hikes starting when I was three.

At the start of each year at the Technification Centre, we would get tested for VO_2 max, recovery, and such things. At the beginning I didn't understand what the results meant, but I learned by talking with my coach. And I was always interested in understanding the reasons behind the workouts—like why she would have me do intervals one day, endurance another day. I'd ask her why it was important to do volume and strength. I was doing a bit more than she told me to, but I respected the interval training and the strength training she gave me. I asked, and she taught.



Kilian Jornet and his sister take a break in the Pyrenees. Photo: Kilian Jornet Collection

When I was seventeen I started university at Font-Romeu in France, taking sports science and studying anatomy and physiology. I learned the theory behind what my coach had been telling me all those years. That's when I started to train myself. In the beginning, I would make a plan for the whole year, mapping out what I would do each week. It worked well and I got good results. But as preparing for the races became more automatic, I stopped planning so far out and in so much detail. I trained more by feel, sensing if I could train more or if I needed rest, if I should push more or less. I had the background to train this way,

based on how I was feeling—a lot of years of training with a coach and the knowledge I'd gained from that.

For me, training in the mountains is a hobby. Sometimes I do things for my sponsors, like traveling and photo shoots, but the training is not my job. I take it seriously, but not so seriously that I lose my love of moving fast in the mountains. ■

Kilian Jornet defines himself as a lover of mountains. He enjoys competition, but above all he conceives of sport as a way to discover the landscape both inside and outside of himself.

